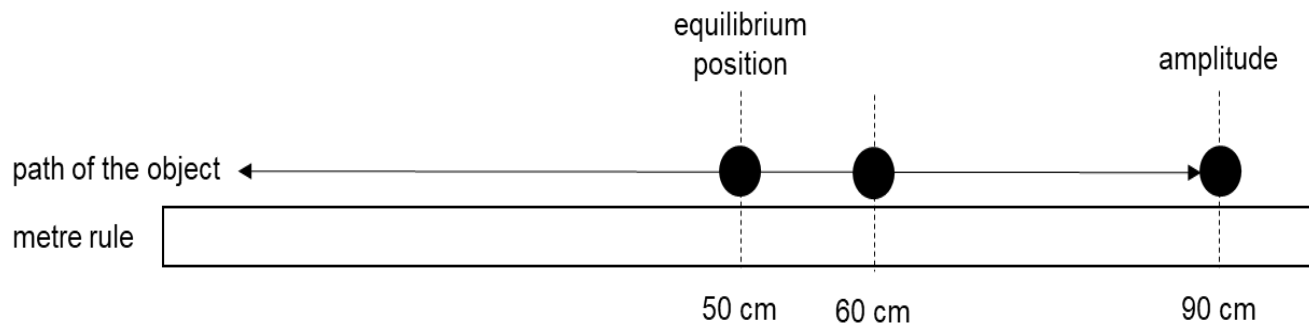


- 14** An object is moving in a horizontal simple harmonic motion parallel to the edge of a metre rule. The equilibrium position of the oscillation and one of the amplitude positions are located at the 50 cm and 90 cm mark of the metre rule respectively as shown.



What is the magnitude of the acceleration of the object at the 60 cm mark in terms of its maximum acceleration a_0 ?

- | | | | | | | | |
|----------|------------|----------|------------|----------|------------|----------|------------|
| A | $0.10 a_0$ | B | $0.25 a_0$ | C | $0.30 a_0$ | D | $0.75 a_0$ |
|----------|------------|----------|------------|----------|------------|----------|------------|

Answer: B

Maximum acceleration $|a_0| = \omega^2 x_0 = \omega^2 (0.90 - 0.50) = 0.40\omega^2$

Acceleration at 60 cm, $a = \omega^2 (0.60 - 0.50) = 0.10\omega^2 = 0.10 \left(\frac{a_0}{0.40} \right) = 0.25 a_0$

- 15** For a vertical oscillating spring-mass system, which of the following changes will **not** decrease